CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 3: Pair of Linear Equations in Two Variables

Previous year board questions, topic-wise: 16 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Graphical method and consistency

1. The graph of a pair of linear equations in two variables is a pair of coincident lines. How many solutions does the pair have? (a) exactly one (b) two (c) none (d) infinitely many CBSE 2026, 1 mark
2. The lines $2x - 3y + 6 = 0$, $2x + 3y - 18 = 0$ and $x = 0$ form a triangle. Draw the graph, write the coordinates of its vertices, and find its area. CBSE 2026, 5 marks
3. A graph shows the lines of two linear equations. The lines are not parallel, but they do not meet in the part of the plane that is drawn. The pair of equations is: (a) consistent, with a unique solution (b) consistent, with infinitely many solutions (c) inconsistent (d) inconsistent, but made consistent by extending the lines CBSE 2024, 1 mark
4. For what value of k does the pair $kx = y + 2$ and $6x = 2y + 3$ have infinitely many solutions? (a) $k = 3$ (b) no such value exists (c) $k = -3$ (d) $k = 4$ CBSE 2023, 1 mark
5. Solve the pair of equations $x = 3$ and $y = -4$ graphically. CBSE 2023, 2 marks
6. Use the graphical method to decide whether the pair $x = 0$ and $y = -7$ is consistent. CBSE 2023, 2 marks
7. Find c if the pair $cx + 3y + (3 - c) = 0$ and $12x + cy - c = 0$ has infinitely many solutions. CBSE 2019, 2 marks

2 Elimination method

8. The pair of equations $2x - 3y + a = 0$ and $2x + 3y - b = 0$ has the solution $x = 1$, $y = 2$. Find a and b, and state how they are related. CBSE 2025, 1 mark
9. Solve the pair $7x - 2y = 5$ and $8x + 7y = 15$, and verify the solution. CBSE 2024, 2 marks
10. $ABCD$ is a rectangle in which $AB = 30$ cm, $DC = (x + y)$ cm, $AD = 14$ cm and $BC = (x - y)$ cm. Find x and y. CBSE 2018, 2 marks

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3 Word problems

11. A person puts money into two schemes, A and B, which pay simple interest at 8% and 9% a year. The interest for one year comes to ₹1,860. If the two amounts had been swapped between the schemes, the interest would have been ₹20 more. How much was put into each scheme? CBSE 2025, 5 marks

12. Three years ago Rashmi's age was three times Nazma's age. Ten years from now Rashmi's age will be twice Nazma's. Find their present ages. CBSE 2024, 3 marks

13. Half the difference of two numbers is 2. The larger number added to twice the smaller number gives 13. Find the two numbers. CBSE 2023, 3 marks

14. A father is three times as old as the sum of the ages of his two children. In 5 years he will be twice as old as the sum of their ages. Find his present age. CBSE 2019, 3 marks

15. A fraction becomes $\frac{1}{3}$ when 2 is taken from its numerator, and becomes $\frac{1}{2}$ when 1 is taken from its denominator. Find the fraction. CBSE 2019, 3 marks

16. A boat covers 30 km upstream and 44 km downstream in 10 hours. In 13 hours it covers 40 km upstream and 55 km downstream. Find the speed of the stream and the speed of the boat in still water. CBSE 2019, 4 marks

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